Oklahoma Induced Seismicity: Causal ETAS Analysis

Bivariate ETAS with Partition Sensitivity

PPDisentangle

2026-03-16

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1 Introduction

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On 18 March 2015 the Oklahoma Corporation Commission (OCC) issued directive **AOI_20150318**, requiring operators within a defined Area of Interest (AOI) in central Oklahoma to reduce wastewater disposal volumes. This report estimates the **all-or-nothing causal effect** of the directive on earthquake rates using the **PPDisentangle** framework.

1.1 Estimand

The target quantity is the **all-or-nothing total treatment effect**: the expected difference in total earthquake counts between a world in which *every* county receives the directive ($\mathbf{Z} = (1, \dots, 1)$) and a world in which *no* county receives it ($\mathbf{Z} = (0, \dots, 0)$):

$$\Delta_{\text{AoN}} = \{\mathbb{E}[N(\mathbf{1})] - \mathbb{E}[N(\mathbf{0})]\},$$

where $N(\mathbf{z})$ denotes the total event count under treatment assignment $\mathbf{z}$.

1.2 Models

The analysis fits four bivariate ETAS models on the **county partition**:

Label	Model	Background	Labels
B	Naive bivariate ETAS	Homogeneous	Location-based
D	SEM bivariate ETAS	Homogeneous	SEM-corrected
E	Naive bivariate ETAS	KDE (non-parametric)	Location-based
F	SEM bivariate ETAS	KDE (non-parametric)	SEM-corrected

Additionally, fits B and D are repeated under alternative **partitioning schemes** to assess sensitivity to spatial resolution.

2 Study Design

Treatment. OCC directive AOI_20150318: mandatory reduction in wastewater disposal volumes within the AOI.

Tessellation. The 77 Oklahoma counties (US Census Bureau via `tigris`). Counties whose centroid falls inside the AOI are treated.

Treated counties (9): Pawnee, Love, Alfalfa, Oklahoma, Logan, Payne, Noble, Grant, Lincoln.

Temporal windows.

- Pre-treatment: 2014-01-01T00:00:00Z to 2015-03-18T00:00:00Z
- Post-treatment: 2015-03-18T00:00:00Z to 2015-06-24T00:00:00Z

Magnitude threshold. $m_0 = 2.5$.

Event counts. - Pre-treatment total: 2472 - Pre-treatment holdout (first 50%): 1236 - Pre-treatment used for estimation/carryover (second 50%): 1236 - Post-treatment: 818 - Total used in SEM estimation (used pre + post): 2054

3 Data

3.1 Magnitude Distribution

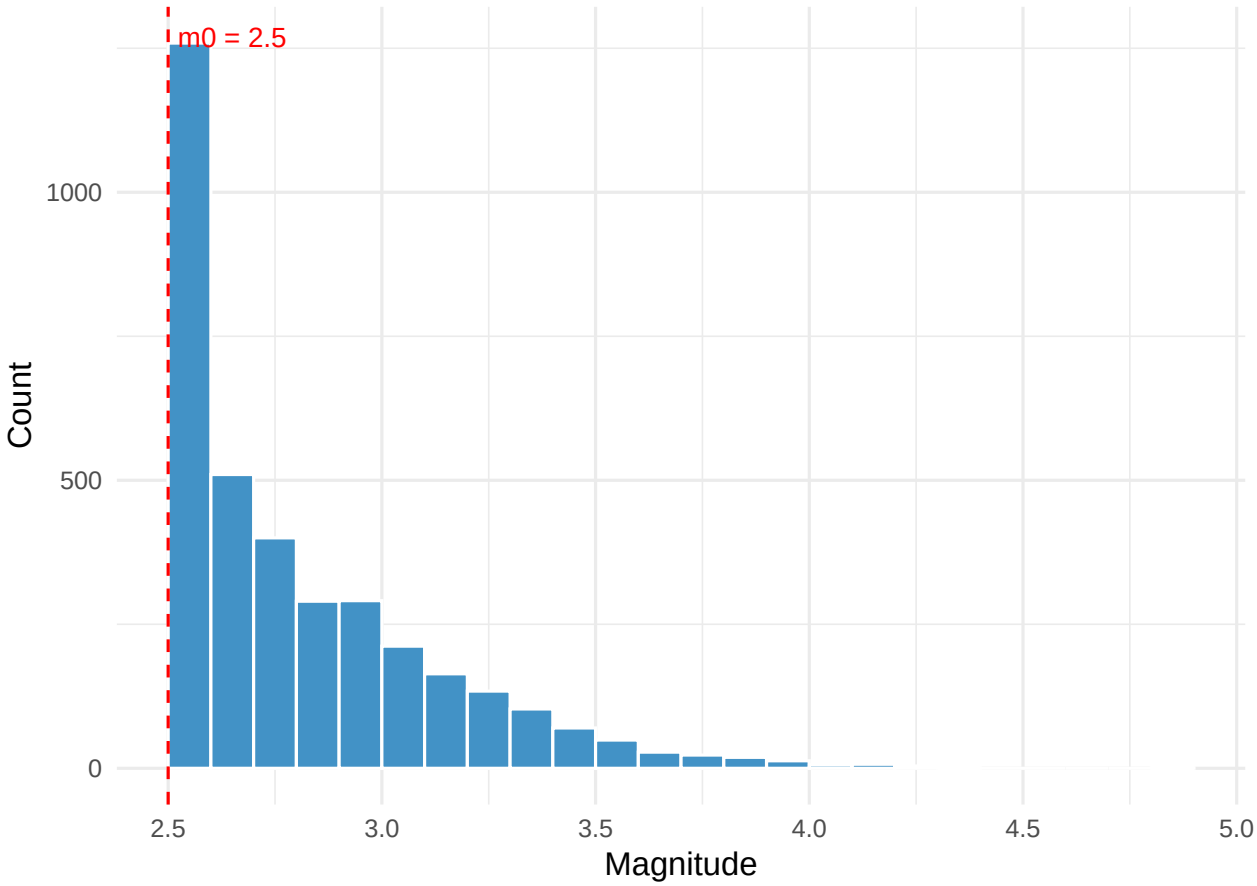


Figure 1: Magnitude distribution of events above the completeness threshold.

3.2 Temporal Profile

3.3 County Partition

3.4 Point Patterns

4 Model

4.1 ETAS Conditional Intensity

$$\lambda(t,x,y) = \frac{\mu}{|S|} + \sum_{j:\, t_j < t} A\, e^{\alpha_m (m_j - m_0)}\, g(t-t_j)\, f(x-x_j, y-y_j \mid m_j)$$

with Omori–Utsu temporal kernel g and power-law spatial kernel f.

4.2 Bivariate Extension

$$\lambda_k(t,x,y) = \frac{\mu_k}{|S_k|} W_k(x,y) + \sum_{l\in\{0,1\}} \sum_{\substack{j:\, t_j < t \\ Z_j = l}} A_{kl}\, e^{\alpha_{m,kl}(m_j - m_0)}\, g(t-t_j)\, f(x-x_j, y-y_j \mid m_j)$$

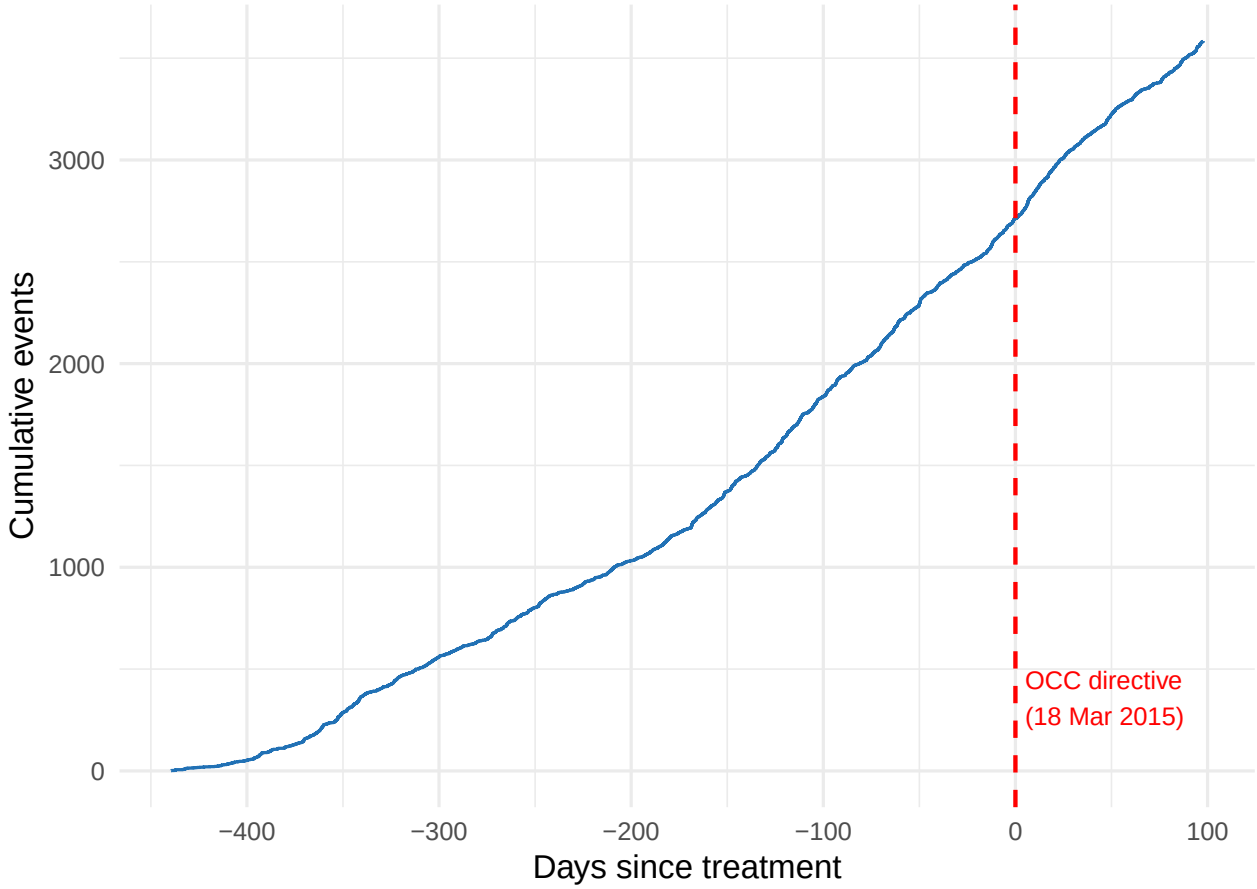


Figure 2: Cumulative event count; dashed line marks the OCC directive.

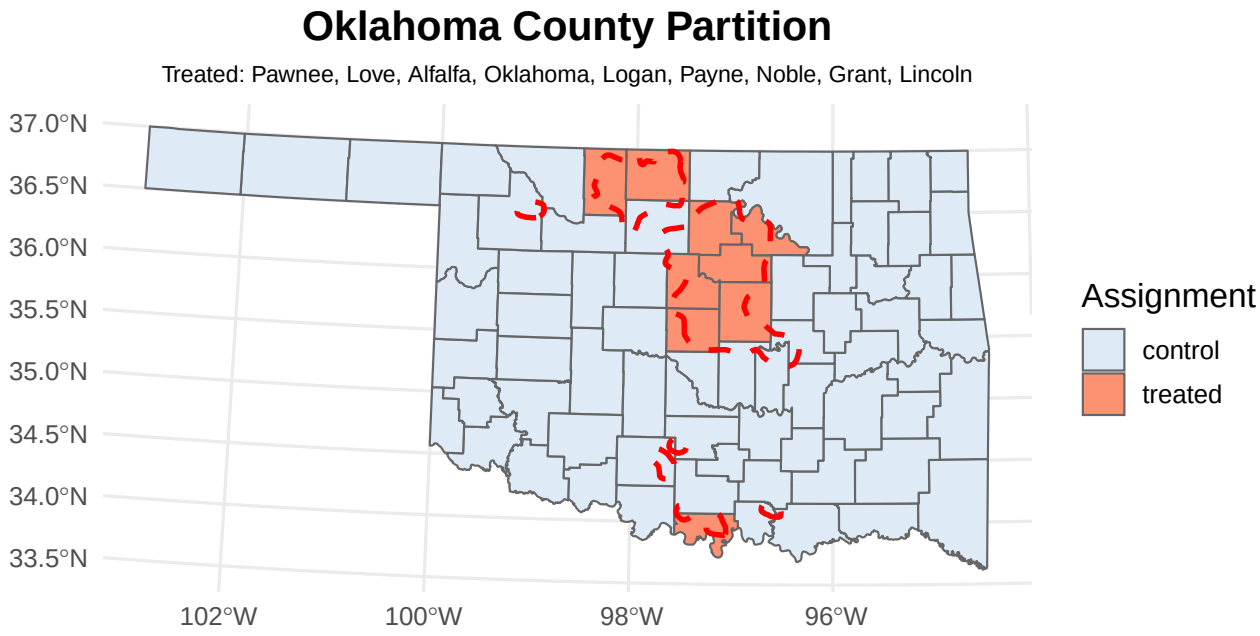


Figure 3: Oklahoma county-level partition. Red counties are treated (centroid inside AOI).

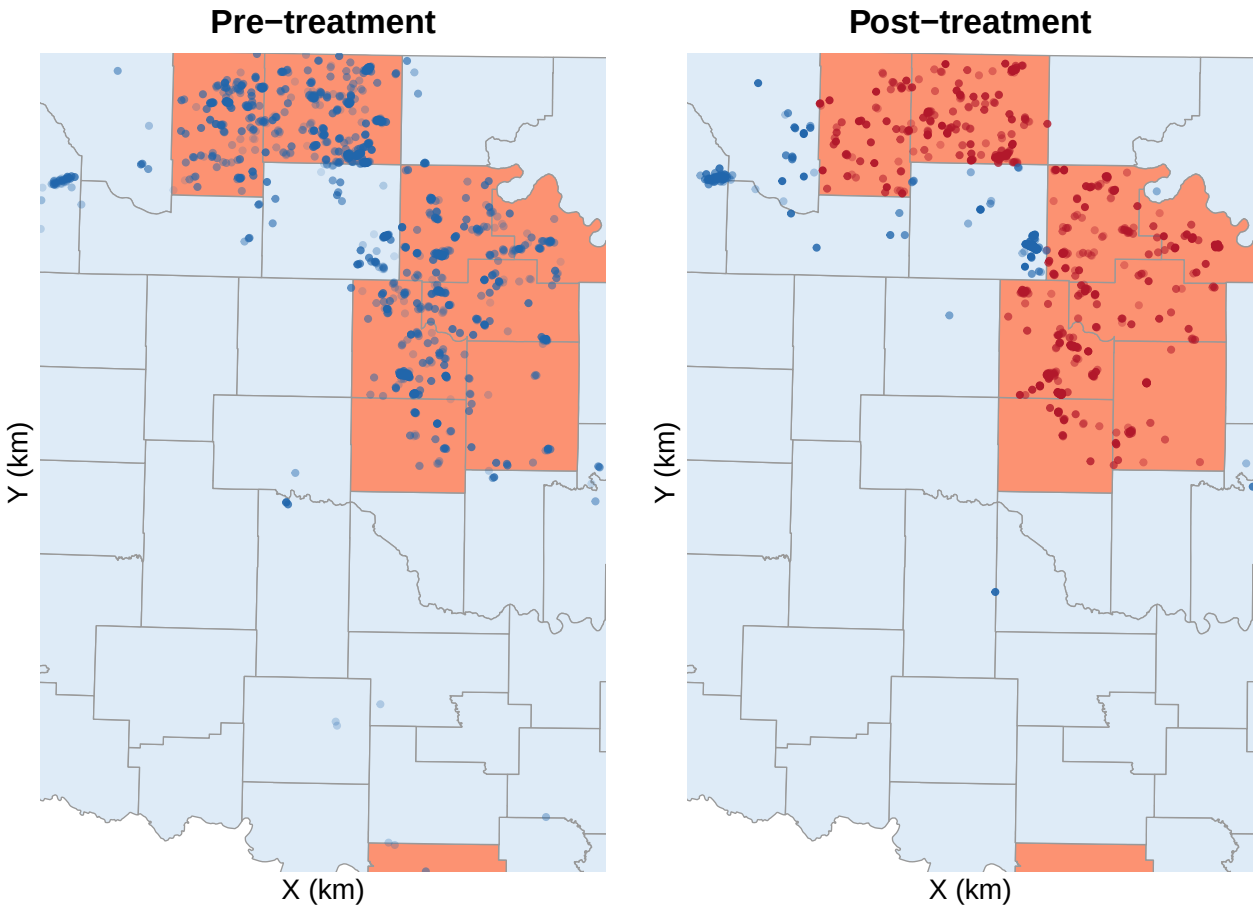


Figure 4: Pre-treatment (left) and post-treatment (right) events overlaid on the county partition.

Off-diagonal terms A_{01} (treated→control) and A_{10} (control→treated) capture cross-excitation. For the KDE background models (E, F), $W_k(x, y)$ is the normalized kernel density estimate from control pre-treatment data; for the homogeneous models (B, D), $W_k \equiv 1$.

4.3 Non-Parametric Background Rate

Table 2: KDE background rate estimation details.

Setting	Value
KDE training events (held-out control pre)	124
Held-out pre-treatment events (all processes)	1236
Pre-treatment events used for estimation	1236
Bandwidth (km)	1.57
Estimated triggering range (km)	4.89
Method	Diggle cross-validation

Only the **second 50%** of pre-treatment events are used for estimation/carryover in both homogeneous and inhomogeneous SEM fits. The **first 50%** is held out for fairness; in the inhomogeneous case, its control events are used to estimate the KDE background via `bw.diggle`.

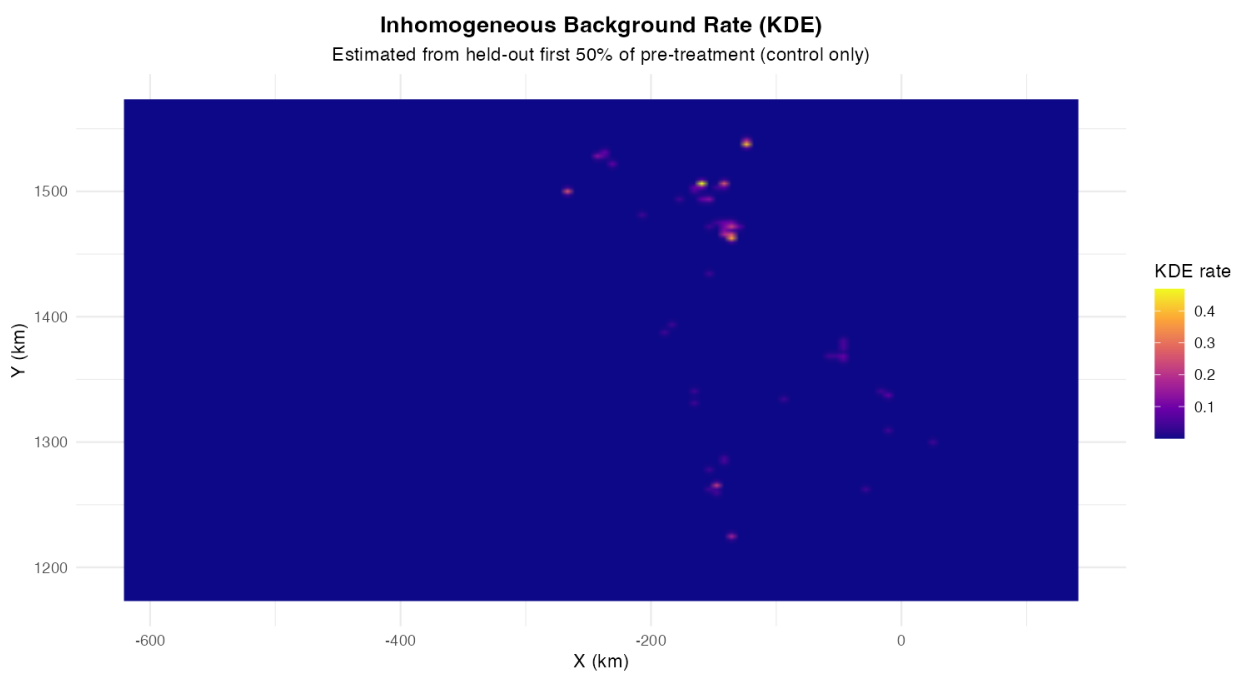


Figure 5: Estimated inhomogeneous background rate from held-out control pre-treatment events.

4.4 Fixed Parameters

Table 3: Fixed structural parameters.

Parameter	Value	Role
c	0.05	Omori onset delay
p	1.20	Omori decay exponent
D	5.00	Spatial scale
	0.50	Mag-spatial scaling
q	1.50	Spatial decay exponent

4.5 SEM Configuration

- **Outer iterations:** 1
- **Inner iterations:** 200
- **SEM inner proposals (`SEM_INNER_PROPS`):** 20
- **SEM labellings (`SEM_N_LABELLINGS`):** 30
- **Change factor:** 0.01

5 Results — County Partition

5.1 Marginal Parameters

Table 4: Marginal ETAS parameters across available bivariate fits.

	B: Naive Biv		D: SEM Biv		E: Naive+KDE		F: SEM+KDE		G: Decode Biv		H: Decode+KDE	
	B:ctrl	B:treat	D:ctrl	D:treat	E:ctrl	E:treat	F:ctrl	F:treat	G:ctrl	G:treat	H:ctrl	H:treat
μ	0.0247	1.3027	0.0299	0.2044	1.8886	0.3697	0.0355	0.2216	0.0000	1.2874	0.0000	1.2701
A	1.1569	0.7536	1.2337	1.0157	0.4953	0.9531	1.0636	1.0081	1.1602	0.7568	1.1336	0.7665
α_m	0.3895	0.8682	0.3794	0.6319	-3.1600	0.8654	0.3429	0.7882	0.4203	0.8977	0.4703	0.8849

5.2 Cross-excitation

Table 5: Cross-excitation parameters for available models.

	B: Naive	D: SEM	E: Naive+KDE	F: SEM+KDE	G: Decode	H: Decode+KDE
A_{01} (treat$\to$ctrl)	0.0031	0.0023	0.0565	0.0076	0.0033	0.0041
$\alpha_{m,01}$	-0.0308	0.5260	-1.0704	0.5805	-0.5888	-1.0772
A_{10} (ctrl$\to$treat)	0.0000	0.0100	0.0000	0.0141	0.0000	0.0000
$\alpha_{m,10}$	-0.3251	0.5633	1.4226	0.5433	3.1879	0.7158

5.3 Parameter Comparison

5.4 SEM Convergence

5.5 SEM Relabelling: Confusion Matrices

5.5.1 Fit D — SEM Bivariate ETAS

Table 6: Fit D: Location vs SEM-inferred label (n = 818 post-treatment events).

Location ↓ / Inferred →	control	treated
control	215 (26.3%)	2 (0.2%)
treated	49 (6%)	552 (67.5%)

5.5.2 Fit F — SEM Bivariate + KDE Background

Table 7: Fit F: Location vs SEM-inferred label (n = 818 post-treatment events).

Location ↓ / Inferred →	control	treated
control	215 (26.3%)	2 (0.2%)
treated	51 (6.2%)	550 (67.2%)

5.5.3 Fit G — Decode Bivariate ETAS

Table 8: Fit G: Location vs Decode-inferred label (n = 818 post-treatment events).

Location ↓ / Inferred →	control	treated
control	207 (25.3%)	10 (1.2%)
treated	4 (0.5%)	597 (73%)

5.5.4 Fit H — Decode Bivariate + KDE Background

Table 9: Fit H: Location vs Decode-inferred label (n = 818 post-treatment events).

Location ↓ / Inferred →	control	treated
control	205 (25.1%)	12 (1.5%)
treated	3 (0.4%)	598 (73.1%)

5.5.5 Relabelling Summary

Table 10: Summary of SEM relabelling across models.

Model	Post events	Ctrl to Treat	Treat to Ctrl	Total relabelled	Pct relabelled
D: SEM Bivariate	818	2	49	51	6.2%
F: SEM Bivariate+KDE	818	2	51	53	6.5%
G: Decode Bivariate	818	10	4	14	1.7%
H: Decode Biv+KDE	818	12	3	15	1.8%

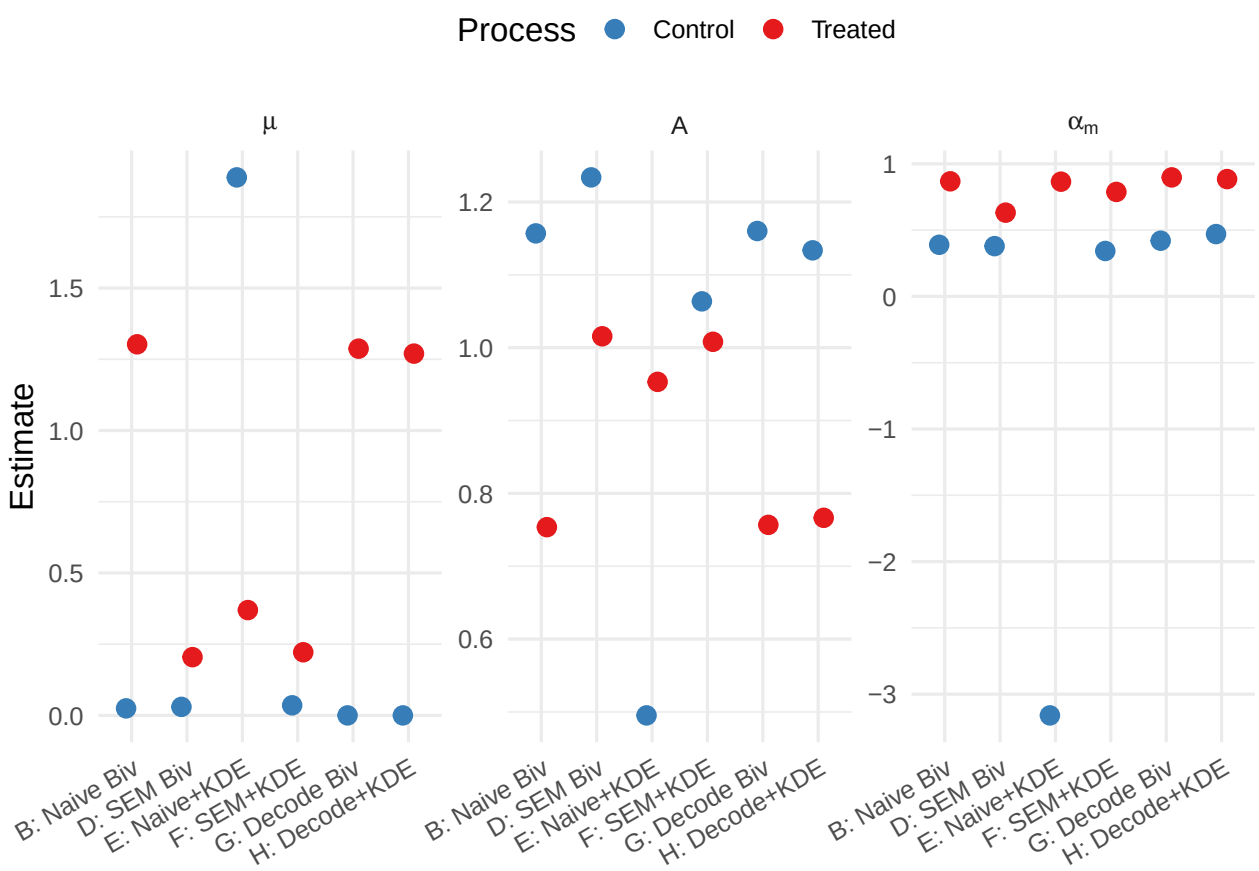


Figure 6: Marginal parameter estimates across bivariate fits.

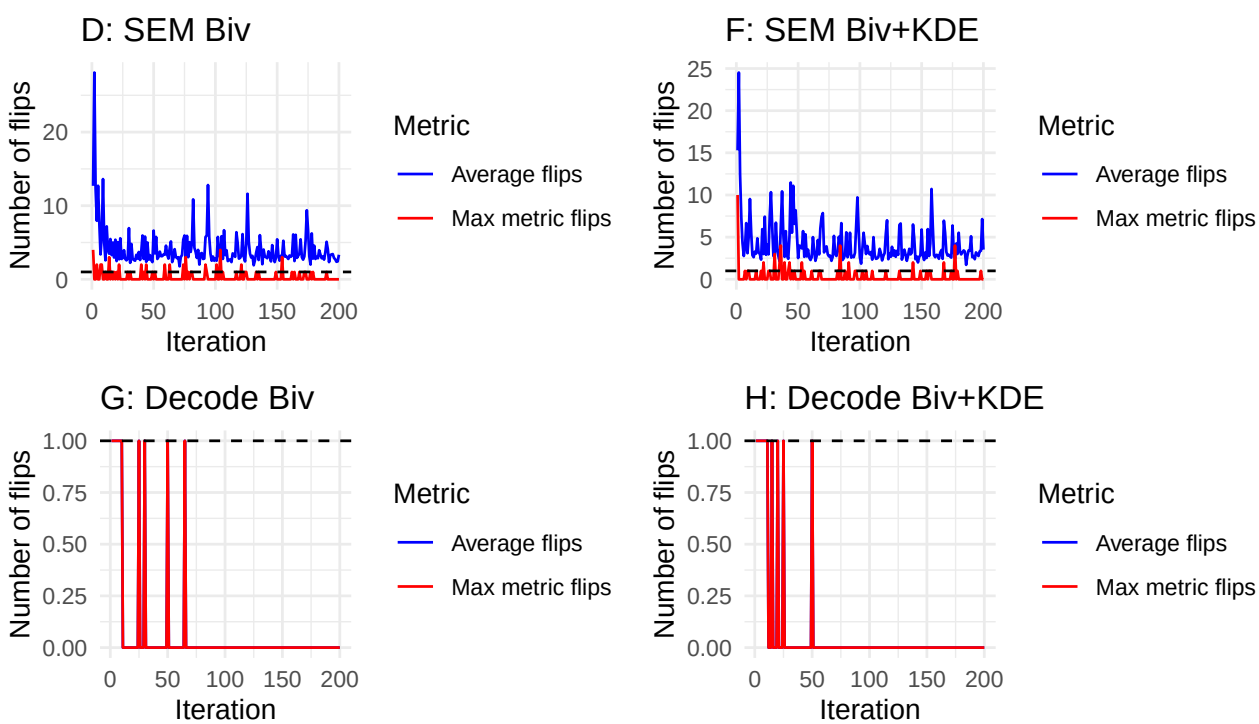


Figure 7: Label flips per relabelling iteration.

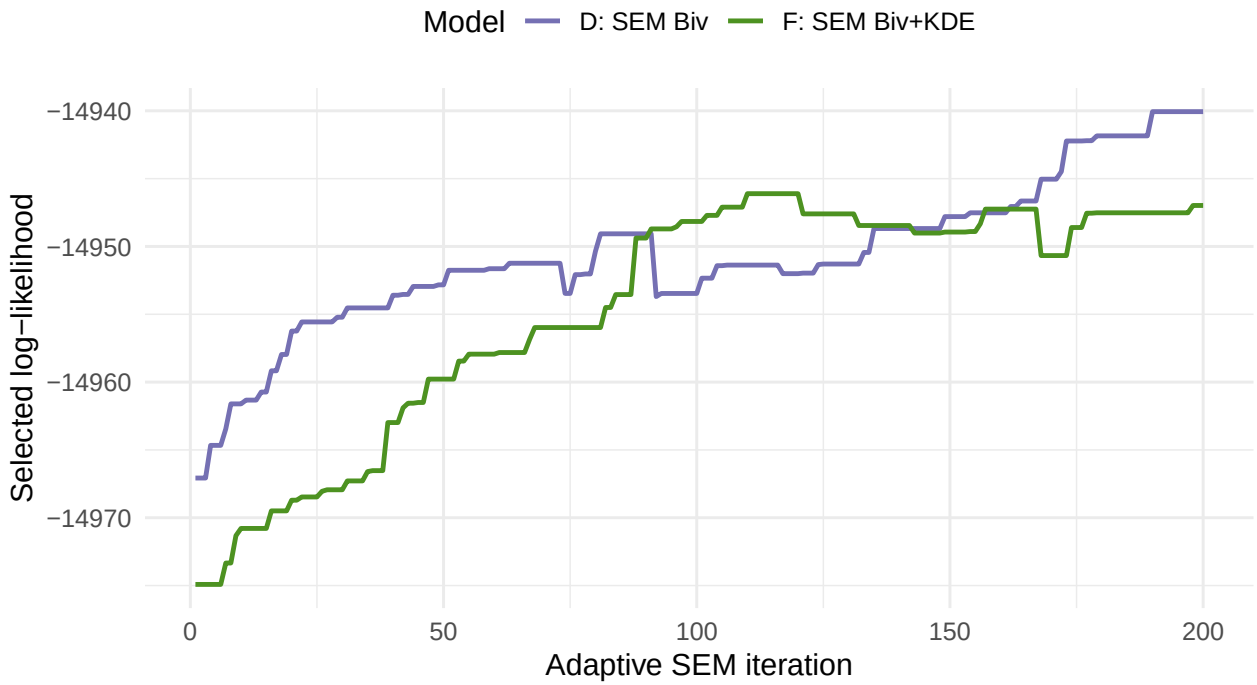


Figure 8: Adaptive SEM selected-label log-likelihood progression (county SEM fits).

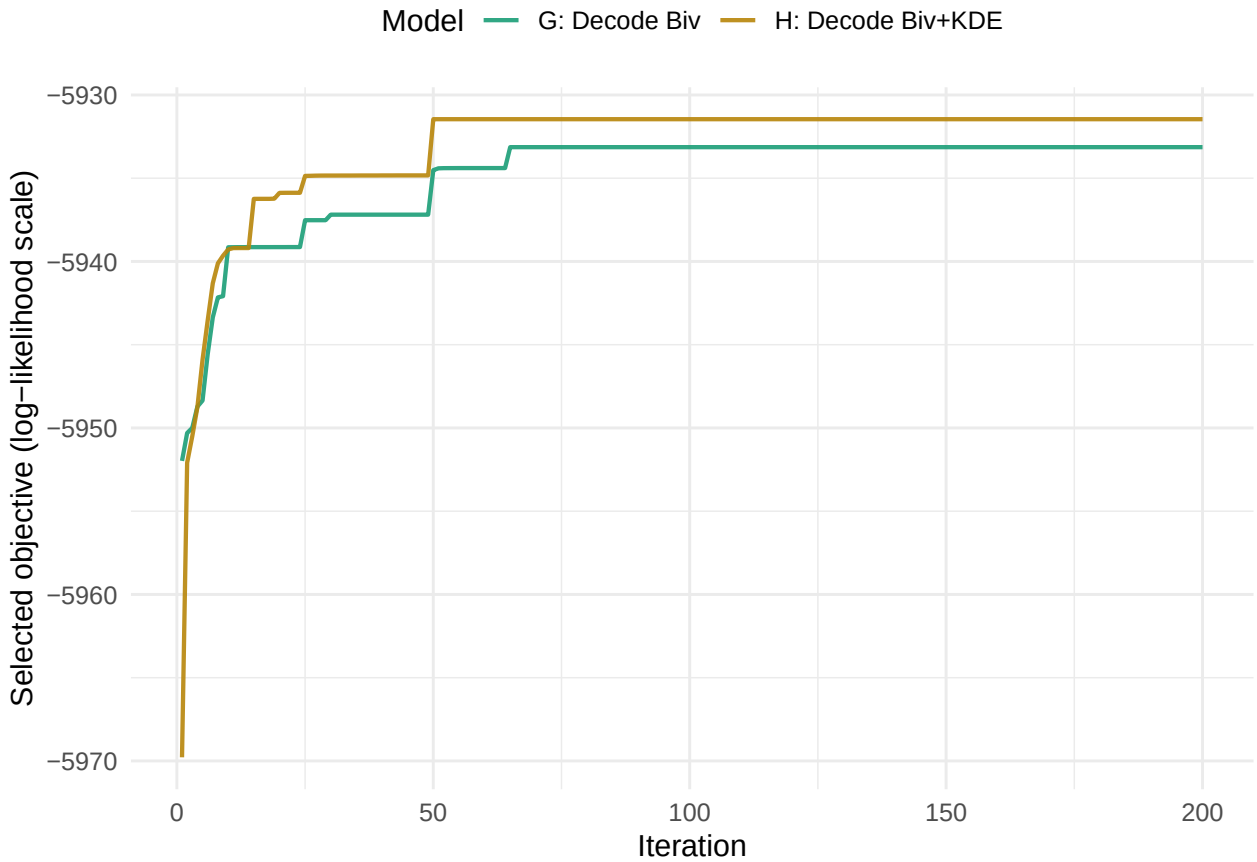


Figure 9: Decode selected log-likelihood progression across relabelling iterations (when Decode models are present).

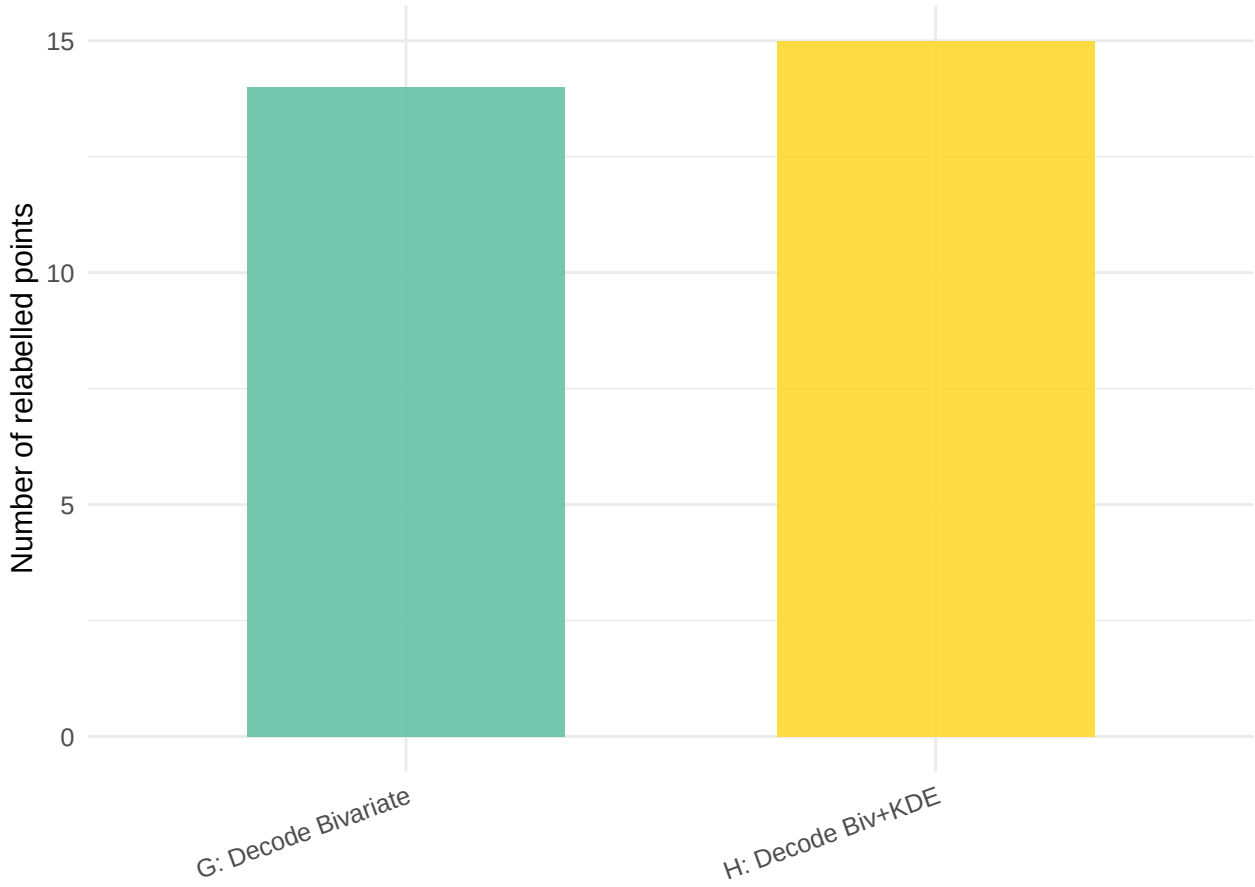


Figure 10: Decode relabelling totals (county models).

6 All-or-Nothing Treatment Effect

6.1 County Partition

Table 11: All-or-nothing total effect over 100 days.

Model	E[N(0)]	E[N(1)]	Total Effect / 77 counties / 100d	SD	Spillover / 100d
B: Naive Bivariate	64	819	755	235	0
D: SEM Bivariate	257	371	114	1165	2
E: Naive Biv+KDE	227	4073	3846	7231	0
F: SEM Biv+KDE	33	4866	4833	5432	2
G: Decode Bivariate	0	837	837	228	11
H: Decode Biv+KDE	0	921	921	203	14

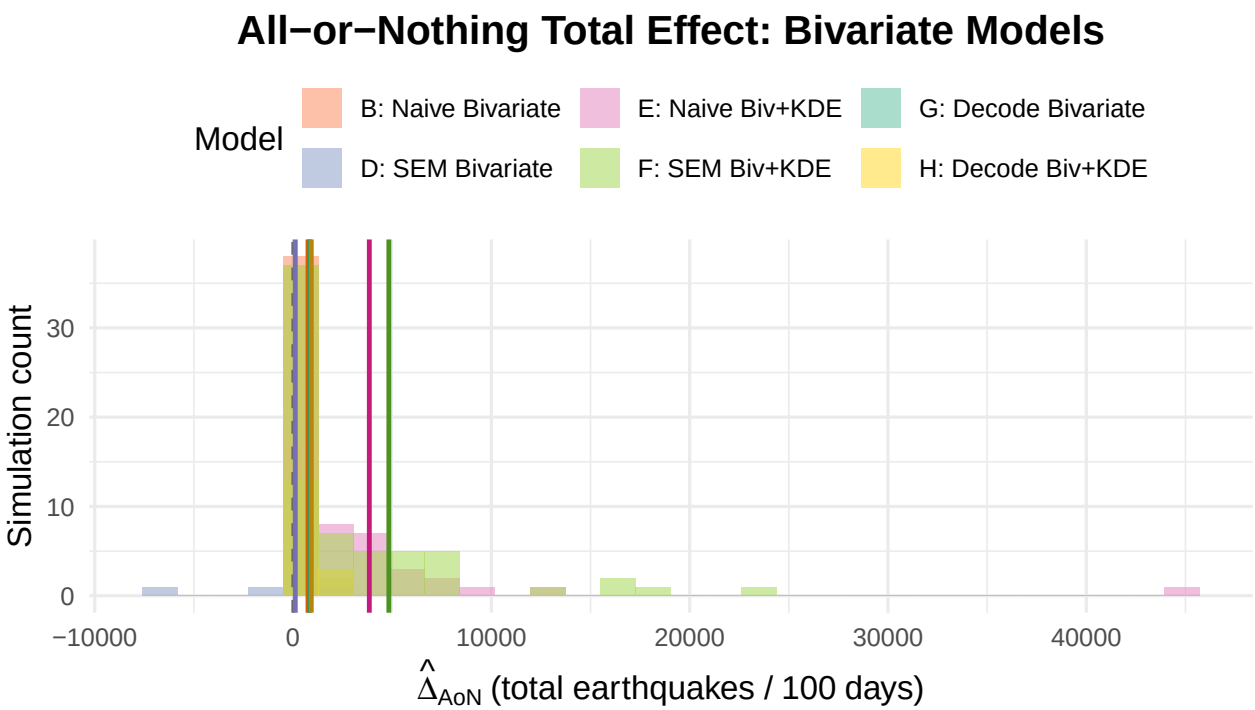


Figure 11: Simulation distribution of the all-or-nothing total effect over the post-treatment horizon.

7 Partition Sensitivity

The analysis is repeated under alternative spatial partitions to assess sensitivity to spatial resolution:

- **Grid partitions** with cell diameters equal to 1, 2, and 5 times the estimated triggering range (4.89 km).
- **AOI region** partition: the AOI polygon defines one treated region, everything else is control (2 tiles total).

For each partition, both the naive bivariate + KDE (E-type) and SEM bivariate + KDE (F-type) models are fitted.

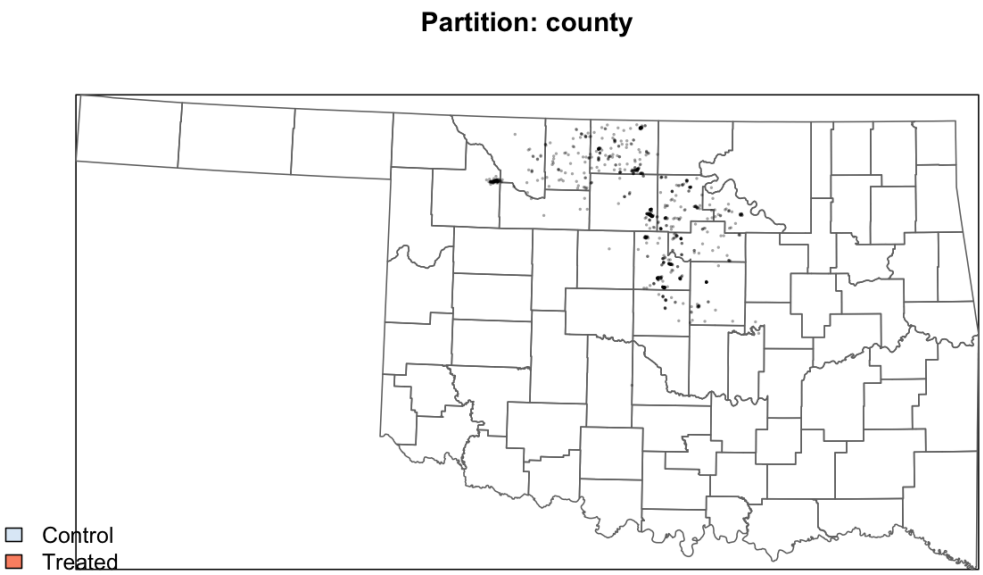


Figure 12: Partition maps for all partition schemes evaluated in the run.

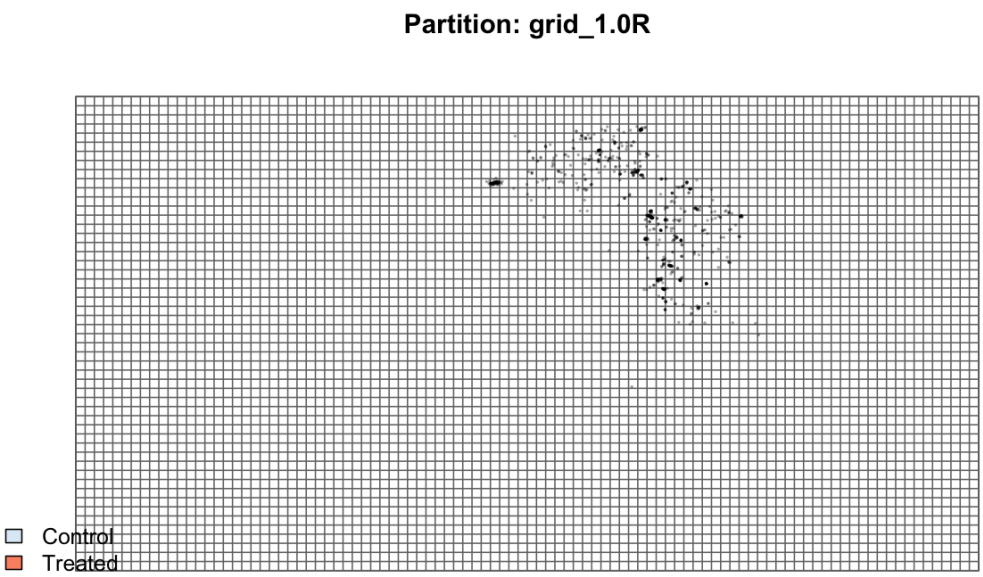


Figure 13: Partition maps for all partition schemes evaluated in the run.

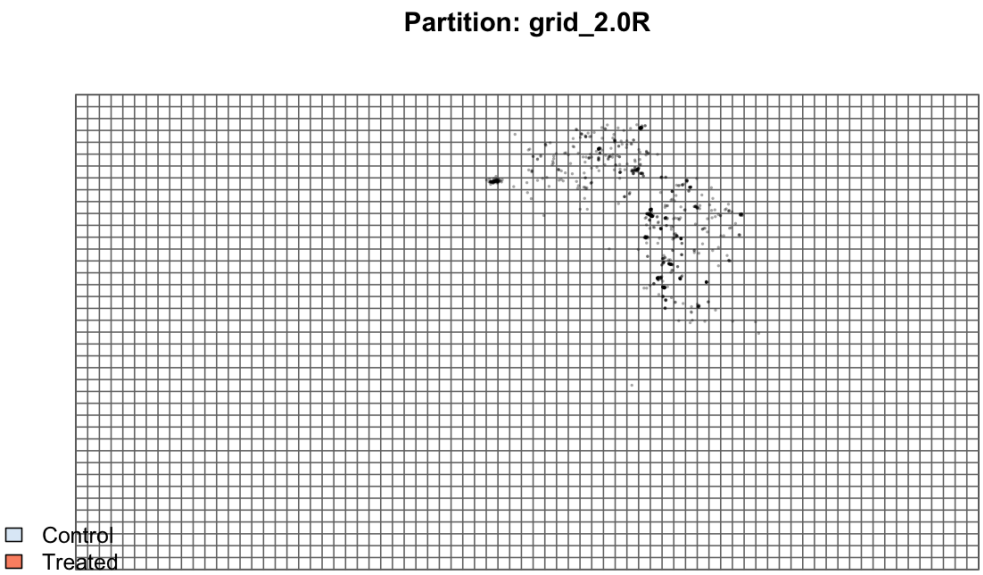


Figure 14: Partition maps for all partition schemes evaluated in the run.

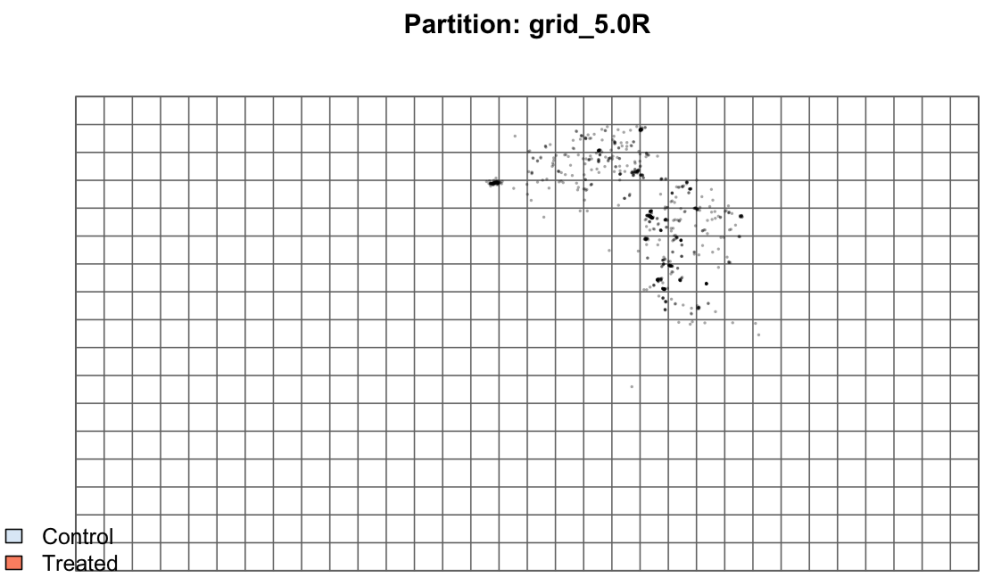


Figure 15: Partition maps for all partition schemes evaluated in the run.

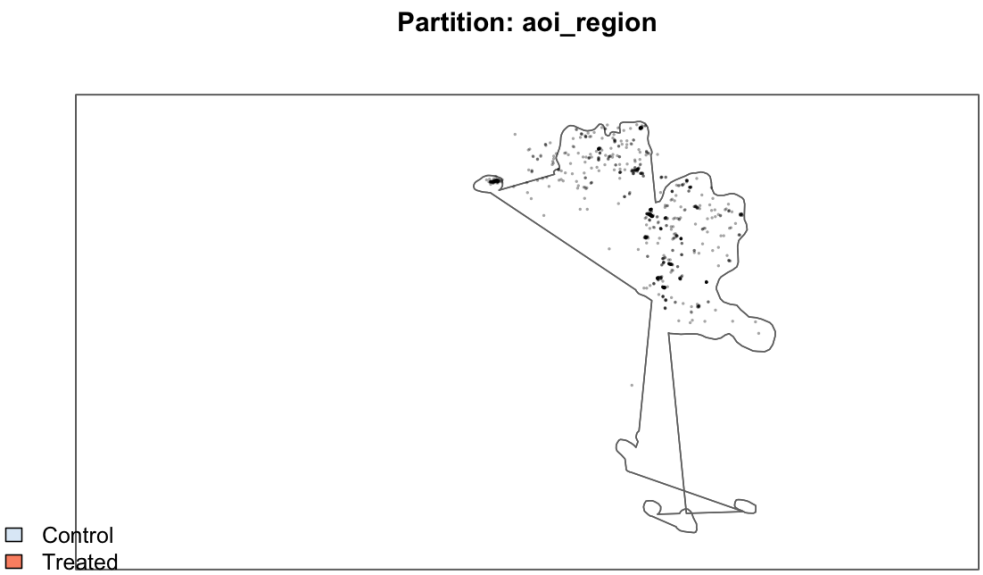
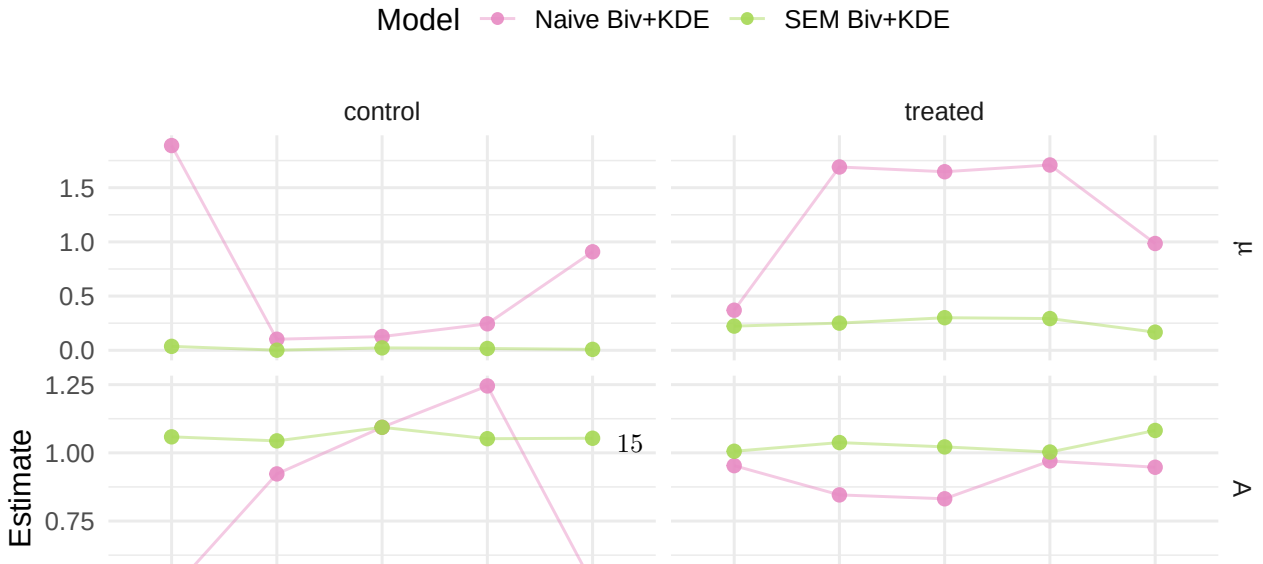


Figure 16: Partition maps for all partition schemes evaluated in the run.

Table 12: Partition comparison over 100 days: E[N(0)], E[N(1)], marginal ETAS paramet

	Partition	Model	Tiles	Treated	Relabelled points	E[N(0)]	E[N(1)]	mu_0	mu_1
1	County	Naive Biv+KDE	77	9	0	227	4073	1.8886	0.3697
2	County	SEM Biv+KDE	77	9	54	33	4866	0.0363	0.2232
9	AOI region	Naive Biv+KDE	2	1	0	46	877	0.1020	1.6910
10	AOI region	SEM Biv+KDE	2	1	62	2	653	0.0011	0.2506
3	Grid (1x triggering range)	Naive Biv+KDE	5096	440	0	29	863	0.1266	1.6471
4	Grid (1x triggering range)	SEM Biv+KDE	5096	440	40	18	586	0.0223	0.3006
5	Grid (2x triggering range)	Naive Biv+KDE	3080	264	0	85	819	0.2444	1.7094
6	Grid (2x triggering range)	SEM Biv+KDE	3080	264	45	12	469	0.0169	0.2927
7	Grid (5x triggering range)	Naive Biv+KDE	544	46	0	159	664	0.9086	0.9850
8	Grid (5x triggering range)	SEM Biv+KDE	544	46	68	8	463	0.0085	0.1672



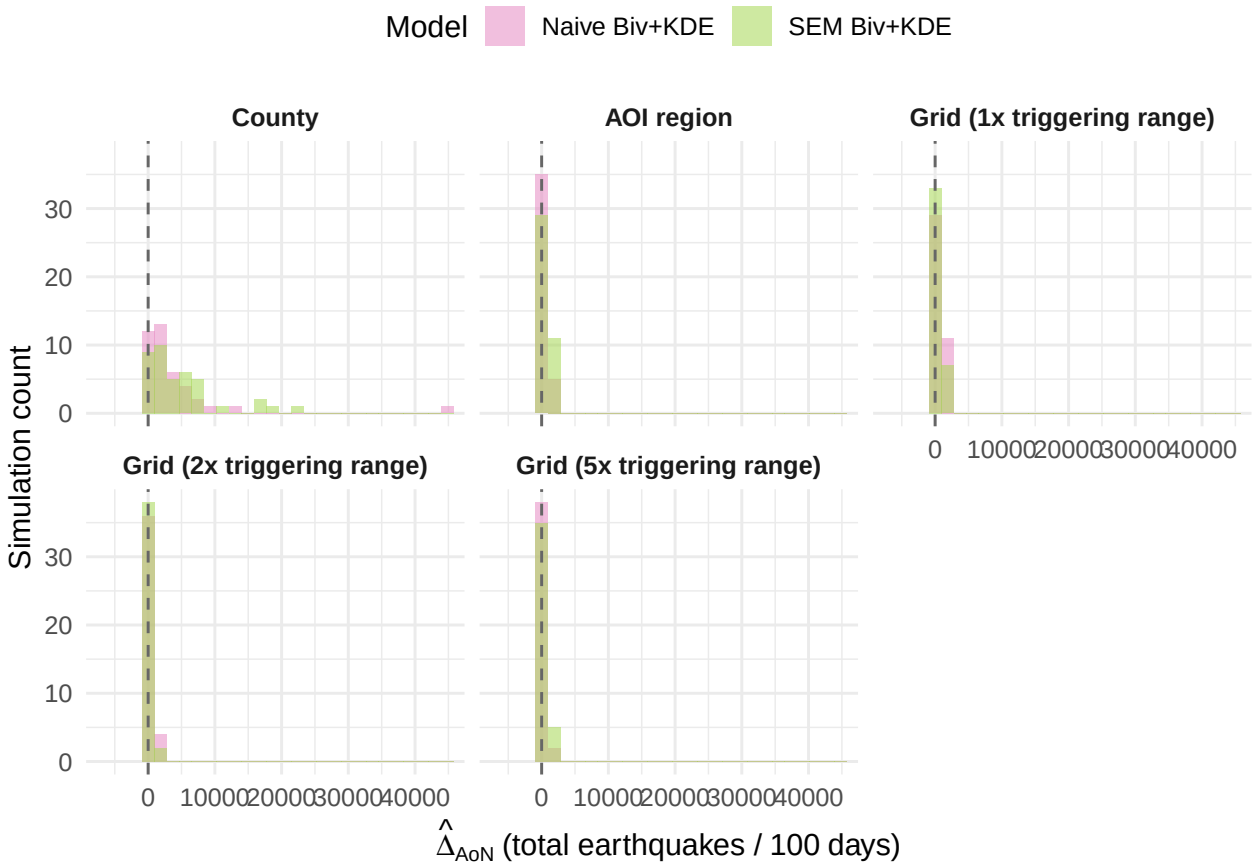


Figure 18: Partition-sensitivity total-effect distributions (same units and x-axis scale as Section 6.1).

7.1 KDE Bandwidth Sensitivity (County Inhomogeneous E/F)

County-level inhomogeneous fits are re-run across KDE bandwidths: Diggle, Diggle$\times2, *Diggle*\times5, *andDiggle*\times$1

Table 13: County inhomogeneous KDE bandwidth sensitivity over 100 days: E[N(0)], E[N(1)] ETAS parameters, and total effect.

Bandwidth	Model	Tiles	Treated	Relabelled points	E[N(0)]	E[N(1)]	mu_0	mu_1	A_0	A_1	al
Diggle	Naive Biv+KDE	77	9	0	211	632	1.8091	0.1642	0.2715	1.0736	
Diggle	SEM Biv+KDE	77	9	38	42	301	0.0368	0.1951	1.0553	1.0034	
Diggle x2	Naive Biv+KDE	77	9	0	218	4008	1.8886	0.3697	0.4953	0.9531	
Diggle x2	SEM Biv+KDE	77	9	44	20	2233	0.0302	0.2104	1.0524	1.0055	
Diggle x5	Naive Biv+KDE	77	9	0	223	1071	1.9120	0.7402	0.1943	0.9535	
Diggle x5	SEM Biv+KDE	77	9	48	22	226	0.0165	0.1921	1.0960	0.9971	
Diggle x10	Naive Biv+KDE	77	9	0	230	684	1.7042	1.2340	0.1468	0.9419	
Diggle x10	SEM Biv+KDE	77	9	32	116	590	0.0372	0.2628	1.0681	1.0129	

8 Interpretation

Inhomogeneous background rate focus. Fits E and F use a spatially inhomogeneous background estimated from the first half of control pre-treatment data. This provides an independent estimate of the baseline seismicity pattern without using post-treatment data.

Background contrast (inhomogeneous only). In Fit E (naive labels), the treated-region background rate μ_1 is 0.2× higher than μ_0. After SEM relabelling (Fit F), this ratio is 6.2×.

All-or-nothing total effect (100-day horizon).

- **Fit E (naive+KDE):** $\hat{\Delta}_{\text{AoN}} =$ **3846** total earthquakes / 100 days
- **Fit F (SEM+KDE):** $\hat{\Delta}_{\text{AoN}} =$ **4833** total earthquakes / 100 days
- **Interpretation:** In the inhomogeneous SEM model (F), treatment looks **worse under treatment**; in the inhomogeneous naive model (E), treatment looks **worse under treatment**.
- **Fit H (Decode+KDE):** $\hat{\Delta}_{\text{AoN}} =$ **921** total earthquakes / 100 days

Partition sensitivity. The alternative partition schemes (grids at different scales and the AOI-as-single-region partition) show how the treatment effect estimate varies with spatial resolution. Coarser partitions aggregate more spatial heterogeneity, while finer grids may better capture localized effects but have fewer events per cell.

9 Configuration

Table 14: Analysis configuration.

Setting	Value
Magnitude threshold	2.5
GR	2.3
Counties (total / treated)	77 / 9
SEM outer iters	1
SEM inner iters	200
SEM inner proposals	20
SEM labellings	30

SEM change factor	0.01
Decode enabled	TRUE
Decode iterations	200
Fitting window (days)	0 – 98
ATE simulation window (days)	100
Simulation replicates	40
Spatial scale D (km)	5
Estimated triggering range (km)	4.89
Grid diameters	4.89 km, 9.77 km, 24.43 km
Test mode	FALSE

10 References

- Ogata, Y. (1988). Statistical models for earthquake occurrences. *JASA*, 83(401), 9–27.
- Zhuang, J., Ogata, Y. & Vere-Jones, D. (2002). Stochastic declustering of space-time earthquake occurrences. *JASA*, 97(458), 369–380.
- Helmstetter, A. & Sornette, D. (2003). Importance of direct and indirect triggered seismicity in the ETAS model of seismicity. *Geophysical Research Letters*, 30(11), 1576.
- Langenbruch, C. & Zoback, M. D. (2016). How will induced seismicity in Oklahoma respond to decreased saltwater injection rates? *Science Advances*, 2(11), e1601542.

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